PLAPPO · INTERNAL RESEARCH BRIEF

Pedagogy, Sources & the Challenge of LLM-Generated Language-Learning Content

What learning science we built on, where it comes from, where it is contested — and an honest account of using a large language model to generate teaching content: what it buys us, how it fails, and what we did about it.

Audience: people I want to walk through what I looked into · **Scope:** the German-learning engine (Plappo) · **Date:** 25 June 2026
Source of truth: `docs/PEDAGOGY_ROADMAP.md`, `docs/DECISIONS.md` (ADR-001 … ADR-023), and the `eval/` engine.

> **How to read this document**
> Three things are deliberately kept separate so you can judge them separately:
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> - **The learning science** — the principles, who established them, and (importantly) where the field genuinely disagrees. Contested claims carry a debated flag rather than being hidden.
> - **What Plappo actually built** on each principle — traced to a specific architectural decision (ADR-xxx) so a claim is checkable in code, not marketing.
> - **The LLM-generation problem** — the long middle of this document. Generating pedagogically-correct language content with an LLM is the riskiest thing we do; it gets a full pros/cons / failure-mode treatment.
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> Honesty rule carried over from the codebase: *label approximations, keep caveats, never claim more than the evals measured.*

Contents

1 · Executive summary

Most language apps optimise for engagement — streaks, taps, recognition. Plappo optimises for the things that actually build a second language: **meaningful input, real production, and focused feedback on a forgetting-proof schedule**, adapting to a *measured* level rather than a self-declared one.

Concretely the loop is: read a level-appropriate German story → answer in your **own written German** → get **focused grammar feedback** (one or two treatable errors, not every mistake) → review the things you got wrong on a **spaced schedule**. Every one of those four verbs maps to a body of research, and every mapping is written down in the codebase so it can be checked.

The honest version: the *core* findings we lean on — retrieval practice, spacing, the generation effect, focused feedback on rule-based errors — are robust and replicated. The famous *headline* versions — Krashen's exact *i+1*, Bloom's exact "2 sigma," "correction always works" — are influential but contested. We build on the solid parts and flag the rest. And the single most fragile component is the one this brief is mostly about: **using an LLM to generate the input**. An LLM told to "write at A2" does not reliably write at A2 — this is now an empirically documented effect, and we reproduced it ourselves. So we never trust the prompt; we gate the output.

> **The one-sentence takeaway for a skeptical reader**
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> The pedagogy is grounded and mostly conservative; the risk is concentrated in LLM-generated content, and the engineering response to that risk is to treat the model as an *untrusted content source* behind a measurable gate — not as a teacher.

2 · The seven principles & their sources

Everything in the product ladders up to seven principles. Each is given here with its grounding and — where it matters — the part the field argues about.

#	Principle	Grounded in & what's contested
P1	**Input before output.** Meaningful, slightly-above-level input (*i+1*); grammar met *in context*, not as isolated rules.	Krashen, *Comprehensible Input*; Long, *Focus on Form* (not Forms). **debated** The *i+1* construct is criticised as not precisely testable — we use it as a design heuristic, not a measurement.
P2	**Production is where acquisition happens.** Producing language forces deeper processing than recognising it.	Swain, *Output Hypothesis*; Schmidt, *Noticing*; Slamecka & Graf, the *Generation Effect*. The generation effect is a robust lab finding.

#	Principle	Grounded in & what's contested
P3	**Feedback must be focused, timely, actionable.** A few treatable errors — not every mistake.	Ferris & Bitchener (focused written corrective feedback); Hattie & Timperley (feed-up / feed-back / feed-forward). **debated** Truscott argued correction is useless; the resolution is "it works *when focused* on treatable, rule-based errors."
P4	**You remember by retrieving on a schedule.** Spaced, interleaved, effortful recall.	Bjork (desirable difficulties); Roediger & Karpicke (testing effect); Cepeda (spacing meta-analysis); **FSRS** as the concrete scheduler. The strongest, least-contested cluster in the whole document.
P5	**Motivation is intrinsic or it's fragile.** Competence, autonomy, relatedness beat points.	Deci & Ryan (Self-Determination Theory; overjustification); Bloom (Mastery); Csíkszentmihályi (Flow). **debated** Bloom's exact 2 sigma is aspirational and contested.
P6	**Adapt to the individual.** 1:1 tutoring is the gold standard; an error profile makes it scalable.	Bloom (2-sigma); Vygotsky (ZPD / dynamic assessment); learner-corpus research. AI-tutoring efficacy is moving fast (2024–2026) — treat any single study as provisional.
P7	**Grammar is the target; vocabulary is the medium.** Train form explicitly; let vocabulary be acquired incidentally through repeated meaningful exposure — capped per level, native in register.	Nation (frequency bands, coverage, incidental learning); Hu & Nation (~98% coverage threshold); Wray (formulaic language); Gilmore (authenticity). The coverage threshold is a strong, empirical anchor.

Why we keep the contested flags visible

It would be easy to cite Krashen and Bloom as settled fact. They are not. Krashen's input hypothesis is enormously influential *and* criticised for being unfalsifiable; Bloom's 2-sigma is famous *and* hard to replicate at that magnitude. Surfacing the disagreement is the point — it tells you which parts of the foundation are load-bearing (P2, P3-focused, P4, P7-coverage) and which are inspirational framing (P1's exact $i+1$, P5's exact 2σ).

3 · From principle to product decision

The link between a research principle and a shipped behaviour is recorded as an Architecture Decision Record. The ones that carry pedagogy:

ADR	Decision	Pedagogical reason & caveat
001 / 003	Validate the feedback engine *before* any UI; headline metric = **false-positive rate on correct sentences**.	**P3.** A tutor that "corrects" good German actively teaches mistakes and destroys trust. Result: **0% false-positive, A1–C2**. "Don't be wrong" was proven first.
002	Schema-constrained JSON for every LLM call.	Parseable, enum-bounded output; the scorer and UI never see a surprise. The foundation that makes everything else measurable.
004 / 007	A single closed **error taxonomy** (20 categories) as the one source of truth.	The model can only emit labels we score — no drift between layers. **caveat** the level→grammar mapping is a *standard-progression heuristic*, not an official CEFR syllabus (none is machine-readable).
008	Real **FSRS-5** spaced repetition (official default parameters; published difficulty/stability equations).	**P4.** Predicts *when* you are about to forget and reviews just in time. Anki's default scheduler since 2023; the highest-leverage retention upgrade.
009	Level is **measured, not set**; grammar and vocabulary tracked as two axes.	**P5/P6.** Self-assessment is unreliable; hand-setting breaks comprehensible input. CEFR explicitly allows jagged profiles. **caveat** "grammar level" isn't an official scale — presented as "≈".
010	Focused feedback: lead with **one** error, collapse the rest — ordered engine-side by pedagogical priority.	**P3.** Ferris/Bitchener: focused correction on treatable errors beats comprehensive correction, which overloads and demotivates.
011 / 012	Don't trust "write at level X" — **gate generated stories** with a different-model judge; over-level words OK *if glossed*.	**P1.** The "+1" of i+1 is supported new vocabulary. This is the central LLM-risk mitigation — see §5.
013	Story content: **generate once, select per user, grow the library**.	A CEFR level is shared across users; regenerating per read re-pays cost and latency forever. Converts a per-read variable cost into a one-time fixed cost, and moves the drift-gate to authoring time where human QA can live.
014	UI: literary serif for content, quiet sans for chrome; one hero action; muted semantics (never alarm-red); animated error→fix diff.	Reduce extraneous cognitive load (Sweller); engineer "noticing" (Schmidt); keep errorful learning safe; avoid the gamification overjustification trap.

ADR	Decision	Pedagogical reason & caveat
018–022	Grammar-first cloze; capped vocabulary with a measurable coverage gate; meaning as an ephemeral gloss; incidental vocab via carrier-word recycling; native register as a second judge axis.	**P7.** The content-design pedagogy — its own section (§4).
023	Analytics are opt-in, pseudonymous, no-PII, with GDPR/BDSG rights.	We can optimise FSRS and calibrate the judge from real learning data without identifying anyone — and learner text sent to the LLM is a separate, documented legal flow.

4 · Content-design pedagogy (P7): simple words, real language

This is where the most opinionated teaching choices live, and where LLM generation does the most work. Four decisions:

4.1 Grammar is the target — the blank is always a grammatical function

Every training item is a cloze whose gap tests a **grammatical decision** (case, adjective/verb ending, verb position, separable prefix, auxiliary, preposition). The cue supplies the *meaning* (the base form), so the only thing produced is the **form**: `Ich helfe ___ Bruder. (mein)` → *meinem* CASE . Grading is exact-match — **no LLM call** — and it feeds the same per-skill grammar profile, so doing review literally raises your measured grammar level.

Why: the moat is German grammatical accuracy, not vocab recall; Schmidt's *Noticing* and the generation effect apply to *form*. (ADR-018)

4.2 Vocabulary is capped per level — and the cap is measured, not vibed

Content draws from a per-level allowed lexicon (cumulative frequency bands; Goethe lists where they exist) plus a small **i+1 budget** of new words that must be glossed. The story gate adds a **deterministic lexical-coverage check**: lemmatise the text (`simplemma` , which handles irregulars like *fuhr→fahren*), band by `wordfreq` Zipf frequency, compute the in-band fraction, reject if unsupported out-of-band content words exceed budget.

Why: reading comprehension needs ~95–98% known-word coverage (Hu & Nation 2000) — the cap is *what makes input comprehensible*, not a limitation. caveat Zipf frequency is a principled but imperfect CEFR proxy, so the threshold stays lenient — the gate catches *gross* drift; the LLM judge handles nuance. (ADR-019)

4.3 Meaning is a subtle comprehension aid, never a trained or featured thing

A word lookup is an ephemeral crutch: a quiet gloss — **no flashcard, no XP, no "added to your deck."** Its only job is to make *this sentence* comprehensible now.

Why: incidental acquisition through meaningful input (Krashen; Webb; Nation), not deliberate paired-associate study. (ADR-020)

4.4 Vocabulary is acquired as a by-product of grammar practice

A gloss-tap is an **"I don't know this word" signal**. Signalled (or never-met) words are preferentially reused as the **carrier words** in upcoming grammar cloze items — so you meet them again, in natural context, while practising grammar. Unknown-status *decays* as you stop looking a word up (several no-lookup encounters → treated as known).

Why: words stick after ~8–12 meaningful encounters (Nation; Webb), not via definitions. **caveat** "didn't look up" ≠ "knows it" — a soft prior over several encounters, never a verdict. (ADR-021)

4.5 The German must be native-authentic *within* the cap

Cap the **range** (which words/structures), never the **naturalness**. Within an A2 lexicon you can write textbook German or the German a native actually uses — we require the latter: real collocations (*eine Entscheidung treffen*, not *machen*), natural connectors and word order, modal particles (*doch, mal, eben*), common contractions (*ins, am, gibt*'s). The story judge scores a separate **naturalness** axis (native vs. textbook/translationese); the gate ships only `naturalness≥4/5` and re-styles on a miss.

Why: native-like fluency is largely formulaic/collocational (Wray), not vocabulary breadth; authentic input beats contrived textbook language (Gilmore). Range and naturalness are independent axes — **simple words, real language**. (ADR-022)

5 · LLM-generated content for pedagogy

This is the part I most want people to understand. An LLM is the only practical way to produce an endless supply of level-appropriate, on-theme, grammatically-targeted German stories. It is also a confident, fluent, and *unreliable* author. The whole engineering story here is: **get the upside, and never let the model be the last word.**

5.1 Where it genuinely helps

Capability	Why it matters pedagogically
Volume of comprehensible input	Acquisition scales with the *volume* of input (the extensive-reading / graded-reader evidence; Day & Bamford; Nation). One story a day isn't enough to move a level. An LLM makes a large graded library economically feasible where hand-authoring does not.
On-demand personalisation	It can weave a learner's specific weak skills and looked-up words into a fresh story (ADR-021) — input that *targets* the detected weakness. That's the move from "we found your gap" to "here's input that closes it" (P6).
Structured, taggable output	Schema-constrained generation (ADR-002) yields a story *plus* its glossary, comprehension questions, a communicative writing task (TBLT), the grammar points it exercises, and a theme — in one call. The tags are what make later cheap selection (ADR-013) possible.
Structured feedback at scale	For the grammar tutor, the model returns error span + taxonomy category + minimal correction + one-line explanation + the skills demonstrated correctly. Validated at **0% false-positive, A1–C2** — the thing that makes the whole loop trustworthy.

5.2 The headline failure: CEFR level drift

> **Telling an LLM "write at A2" does not reliably produce A2**
>
> This is now empirically documented — "*Alignment Drift in CEFR-prompted LLMs for Interactive Spanish Tutoring,*" arXiv 2505.08351 (2025): models measurably drift off a prompted CEFR level over a conversation. **We reproduced it ourselves** — a longer-story sweep leaked over-level words into A2 stories. The conclusion baked into the architecture (ADR-011): **do not trust the generation prompt**; verify the output.

Drift is insidious because the output still *looks* good — it's fluent, on-topic, plausibly leveled. A human skimming it would likely approve it. The leak is a handful of B1/B2 content words inside an A2 story, which is exactly enough to push the text below the ~98% known-word coverage threshold and break comprehensibility for the actual A2 learner. The failure is invisible to vibes and only visible to measurement.

5.3 A catalogue of failure modes (and the mitigation for each)

Failure mode	What it looks like	Mitigation in Plappo
Level drift (too hard)	Over-level vocab/grammar leaks into a story for a lower level; breaks *i+1* comprehensibility.	Different-model judge classifies above/on/below + a deterministic in-band coverage gate; regenerate steering away from the offending words (ADR-011/012/019).
Level drift (too easy)	A "B1" story that's really A2 — no acquisition, wasted read.	The judge's three-way classification flags *below*, not just above.
Stilted "Lehrbuchdeutsch"	Technically in-level but reads as translated-from-English textbook German; teaches unnatural phrasing.	A separate **naturalness** judge axis (1–5) + `unnatural_phrases` flags; ship only ≥4, otherwise re-style (ADR-022).
Inventing errors in correct German (feedback side)	The tutor "corrects" a sentence that was already right — the single most trust-destroying failure.	Headline metric is the *false-positive rate on correct sentences*; ~half the eval set is already-correct; explicit "do NOT rewrite correct German" rule (ADR-003).
Self-agreement / grading itself	Using the same model to write and to judge — it rubber-stamps its own drift.	Generate with one model, judge with a *different* one (ADR-011).
Judge is not ground truth	The "stays-in-level" checker is itself an LLM — a signal, not a certified CEFR grade.	Acknowledged explicitly; backed by a *deterministic* coverage check so the gate doesn't rest on the LLM alone; calibration against human ratings is an open task.
Hallucinated glossary / wrong lemma	A gloss with the wrong part of speech or a lemma that doesn't match the surface form silently corrupts the coverage check and the carrier-word loop.	Schema requires surface form *exactly as it appears* + lemma + POS; coverage lemmatises independently with `simplemma` rather than trusting the model's lemma.
Cost / latency in the request path	Generating + gating on every read re-pays the cost forever and adds latency every time.	Generate-once / select-per-user library; most reads hit the corpus, not a generation call (ADR-013/016).
Subtle factual / cultural errors	Plausible-but-wrong facts in a "nature" or "city" story the learner absorbs as true.	Partly mitigated by short, simple, low-stakes content; *not* fully solved — flagged as residual risk (§7).

5.4 Pros and cons, side by side

What LLM generation buys us

- Effectively unlimited graded input → the **volume** extensive reading needs.
- Real personalisation: weak-skill targeting and looked-up-word recycling baked into fresh stories.
- One structured call yields story + glossary + questions + communicative task + tags.
- Trustworthy structured grammar feedback (0% false-positive, A1–C2).
- Native-register prose is achievable *within* a capped lexicon — "simple words, real language."
- Cost collapses to near-zero when amortised via the shared library.

What it costs / risks

- Confidently off-level output that *looks* fine (the drift problem).
- Every gate is itself probabilistic — a judge LLM is a signal, not truth.
- Naturalness and factual accuracy are not guaranteed by the prompt.
- Validation is ongoing work, not a one-time pass — models change under you.
- Automated feedback is real but **weaker than a human teacher** (§5.6).
- Privacy/legal surface: learner text goes to a third-party processor (ADR-023).

5.5 What we built to contain it — the gating architecture

The design principle is to treat the model as an **untrusted content source** and wrap it in checks that get progressively cheaper and more deterministic:

1. **Constrain the output shape** — schema-enforced JSON so the result is always parseable and tag-bounded (ADR-002).
2. **Judge with a different model** — level classification (above/on/below) + a naturalness score, by a model that didn't write the text (ADR-011).
3. **Add a deterministic backstop** — lemmatise + frequency-band the actual words and compute in-band coverage; this needs no model and can't be talked into agreeing with itself (ADR-019).
4. **Apply the i+1 exception** — over-level words are allowed *only* if the story glosses them (supported new vocabulary is the legitimate "+1") (ADR-012).
5. **Regenerate with steering, then give up** — on a miss, re-prompt away from the offending words and stilted phrasings, bounded by `max_tries`; a story that never passes is simply not shipped.
6. **Move the gate to authoring time** — because content is generated into a shared library, the expensive rigorous gate (Pro model, full datasets) runs *offline*, where human QA can also sit, and runtime just selects (ADR-013/016).

> **The pattern, stated generally**
>
> For pedagogical LLM content: **generate cheap, verify expensive, and keep at least one verifier that is not itself an LLM.** The deterministic coverage check is the load-bearing one — it's the only part of the pipeline that cannot be charmed by fluent output.

5.6 The honest ceiling: automated feedback is not a teacher

> A meta-analysis of feedback channels finds teacher online feedback (g ≈ 2.25) ≫ automated (g ≈ 0.70) ≫ peer. Automated feedback — what we ship — **helps, but is not as strong as a good human teacher.** The honest framing for Plappo is therefore *scalable practice between lessons*, not a teacher replacement. A separate 2025 meta-analysis also finds near-zero benefit of LLM tools to *self-regulation* over short studies — so we must not assume the tool builds study habits; we design for them (FSRS cadence, reflection) and measure over months.

6 · How we keep ourselves honest — the measurement plan

The project's spine is "prove the risky thing with an eval *first*, then build UI on top." The signals we hold ourselves to:

Signal we watch	What it proves
Engine false-positive rate stays ~0% as content scales	P3 — trust (the make-or-break metric)
Errors in a skill *decline* over spaced repetitions (not just first-try accuracy)	P4 — durable retention, not cramming
Time-to-level-up correlates with real CEFR can-do performance	P1 / P5 — valid progression, not a vanity number
Generated stories stay in-band on coverage *and* naturalness as the library grows	§5 — drift stays contained at scale
Retention driven by competence + relatedness, not streak pressure	P5 — intrinsic motivation
Profile-selected content reduces repeat errors faster than random content	P6 — personalisation actually works

Calibration tasks already wired into `analyze_events.py` (from consented logs): per-user FSRS parameter optimisation; calibrating the story judge against human CEFR ratings; tuning the Zipf bands; validating the level→grammar map against a real Goethe syllabus.

7 · Open risks we have *not* closed

Risk	Why it matters	Status
Story generation staying in-level at scale	The input half of the app; empirically a real LLM failure (arXiv 2505.08351, reproduced).	Contained by the gate; needs ongoing eval as the library and models change — never "done."
Corrective-feedback dosage	Over-correct → demotivate (violates P3); under-correct → stall.	Tunable via the focused-feedback ranking; not yet tuned on real learner data.
Advanced grammar is partly style, not rules	At C1/C2 much "grammar" is register/preference; the engine should stay silent, not invent errors.	Handled by testing C1/C2 via correct-sentence (false-positive) cases.
No canonical machine-readable CEFR grammar syllabus	Our level→skill map is *standard*, not authoritative.	Validate against Goethe inventories before any high-stakes claim.
Judge ≠ ground truth; factual/cultural slips	The level/naturalness judge is an LLM; generated facts can be plausibly wrong.	Deterministic coverage backstop helps level; factual accuracy is only partly mitigated — residual.

Risk	Why it matters	Status
Self-regulation & relatedness	LLM tools show near-zero self-regulation benefit short-term; relatedness is the missing leg of intrinsic motivation.	Design for habit (FSRS, reflection) and measure over months; relatedness features are later-phase.

8 · Annotated bibliography

Every claim above is traceable here. **free** = openly available; **debated** = checking the source will (correctly) show an active scholarly disagreement, not a settled fact. Verify exact volume/page/DOI before quoting in anything public.

P1 — Comprehensible input / Focus on Form

- Krashen, S. (1982) *Principles and Practice in Second Language Acquisition*; (1985) *The Input Hypothesis*. **free** (sdkrashen.com) **debated** — the *i+1* construct is criticised as not precisely testable.
- Long, M. (1991) "Focus on form: A design feature in language teaching methodology"; expanded in Long (2015).
- Nation, I.S.P. (2009) *Teaching ESL/EFL Reading and Writing*; Day & Bamford (1998) *Extensive Reading in the Second Language Classroom* — the extensive-reading evidence base.

P2 — Output, Noticing, Generation

- Swain, M. (1985) "Communicative competence: some roles of comprehensible input and comprehensible output" (Output Hypothesis).
- Schmidt, R. (1990) "The role of consciousness in second language learning," *Applied Linguistics* 11(2), 129–158 (Noticing).
- Slamecka, N. & Graf, P. (1978) "The generation effect: delineation of a phenomenon," *J. Exp. Psychology*.

P3 — Focused corrective feedback

- Truscott, J. (1996) "The case against grammar correction in L2 writing classes," *Language Learning* 46(2), 327–369, DOI 10.1111/j.1467-1770.1996.tb01238.x. **debated** — read the *against* side first.
- Ferris, D. (2011) *Treatment of Error in Second Language Student Writing* (2nd ed.); Bitchener & Ferris (2012) *Written Corrective Feedback in SLA and Writing* — the *for, but focused* side.
- Hattie, J. & Timperley, H. (2007) "The power of feedback," *Review of Educational Research* 77(1), 81–112, DOI 10.3102/003465430298487.
- Chen & Renandya (2020) meta-analysis, *TESL-EJ* 24(3) — overall **g ≈ 0.59** across 35 studies. **debated** — weight of evidence favours *focused* CF on *treatable* errors.

P4 — Spaced retrieval, interleaving, scheduling

- Roediger & Karpicke (2006) "Test-enhanced learning," *Psychological Science* 17(3), 249–255 (testing effect).
- Bjork & Bjork (2011) "Making things hard on yourself, but in a good way" (desirable difficulties).
- Cepeda et al. (2006) "Distributed practice in verbal recall tasks," *Psychological Bulletin* 132(3) (spacing meta-analysis).
- **FSRS** — open-spaced-repetition community (Jarrett Ye et al.); Anki's default since v23.10 (Nov 2023), replacing SM-2; DSR memory model. **free** github.com/open-spaced-repetition.

P5 — Motivation, mastery, flow

- Deci & Ryan (2000) "The 'what' and 'why' of goal pursuits," *Psychological Inquiry* 11(4), 227–268. **free** selfdeterminationtheory.org (autonomy/competence/relatedness; overjustification).
- Bloom, B. (1984) "The 2 Sigma Problem," *Educational Researcher* 13(6), 4–16. **debated** — the 2σ figure is aspirational; exact replicability contested.
- Csíkszentmihályi, M. (1990) *Flow: The Psychology of Optimal Experience*.

P6 — Adapt to the individual

- Vygotsky, L. (1978) *Mind in Society* (Zone of Proximal Development).
- Poehner & Lantolf (2005) "Dynamic assessment in the language classroom," *Language Teaching Research*.

- Ellis, R. (2003) *Task-based Language Learning and Teaching* (TBLT).

P7 — Vocabulary, coverage, formulaic language, authenticity

- Nation, I.S.P. (2013) *Learning Vocabulary in Another Language* (2nd ed.) — frequency bands, coverage, encounters needed.
- Hu & Nation (2000) "Unknown vocabulary density and reading comprehension," *RFL* 13(1). **free** — the ~98% coverage threshold (the empirical basis for capping vocabulary).
- Webb, S. (2007) "The effects of repetition on vocabulary knowledge," *Applied Linguistics* 28(1).
- Laufer & Nation (1995) "Vocabulary size and use," *Applied Linguistics* 16(3) — lexical frequency profiling (the measurable cap).
- Wray, A. (2002) *Formulaic Language and the Lexicon* — native-like fluency is chunk-based.
- Gilmore, A. (2007) "Authentic materials and authenticity in foreign language learning," *Language Teaching* 40(2). **debated** .

Standards & the 2024–2026 AI/LLM evidence (re-check yearly)

- **CEFR Companion Volume (2020)**, Council of Europe. **free** coe.int — authoritative level descriptors / can-do statements.
- **LLMs drift off the target CEFR level** — "Alignment Drift in CEFR-prompted LLMs," arXiv 2505.08351 (2025). **our biggest risk** — validates building a "stays-in-level" checker rather than trusting the prompt.
- **AI in language teaching — positive across skills:** *Computers & Education: AI* (2025), ScienceDirect S2666920X25001626 — positive but large variation by level/design.
- **LLMs in education — systematic reviews (2025):** ScienceDirect S2666920X25001699; Springer 10.1007/s43621-025-01094-z (personalised learning).
- **Self-regulation caution:** meta-analysis of LLM effects, arXiv 2509.22725 (2025) — *near-zero* self-regulation benefit (studies ≤10 weeks).
- **Online feedback meta-analysis:** teacher (g ≈ 2.25) ≫ automated (g ≈ 0.70) ≫ peer — *Asia-Pacific Education Researcher*, Springer 10.1007/s40299-021-00594-6.
- **Recent feedback synthesis (2024):** "How effective is feedback for L1, L2, and FL learners' writing? A meta-analysis," ScienceDirect S0959475224000884.

> **Honesty note (carried verbatim from the roadmap)**
>
> The *core* findings — retrieval practice, spacing, output, focused feedback on rule-based errors — are robust and replicated. The headline *figures* and *strong* versions — Krashen's *i+1*, Bloom's exact 2σ, "correction always works" — are influential but debated. That's precisely why the sources are listed: so the reader checks them and builds on the solid parts. Core SLA is stable; FSRS internals, CEFR editions, and AI-tutoring evidence evolve — revisit yearly.